

ETF Risk-Signal Diversification

Operations Research–Oriented Journal Article Blueprint, Code Audit, and Implementation Report

Dynamic Long-Only ETF Allocation with a Causal Universe, Posterior-Predictive CVaR, Backward/Forward Risk-Signal Diversification, and Stable Walk-Forward Selection

Prepared from the uploaded portfolio code package, prior article blueprint, implementation report, and interactive dashboard reference

Date: 2026-07-21

Research-use warning

The uploaded archive did not contain the raw point-in-time ETF dataset. The historical strategy was therefore not rerun in this environment. The code was audited, corrected, compiled, unit-tested, and smoke-tested on synthetic reporting data. No strategy is declared superior until the corrected pipeline is executed on the real dataset under the frozen out-of-sample protocol.

Executive Summary

This report serves two purposes. First, it is a complete writing plan for a potential Operations Research manuscript. Second, it documents a forensic audit and redesign of the uploaded Python framework so that the journal narrative and the executable decision policy match. The recommended paper is not a story about a backtest that happened to win. It is a dynamic stochastic-optimization paper about how a decision maker constructs a changing investable ETF universe, generates horizon-consistent return scenarios, solves a coherent tail-risk-budget problem, executes only part of the target change, and controls total risky exposure using complementary backward and forward risk signals.

The strongest defensible contribution is model consistency. The same posterior-predictive scenario fan that defines the CVaR allocation budget also produces a forward model-CVaR exposure timer. This forward timer is combined with a backward realized-volatility/GARCH timer and a bounded conviction timer based on fractional Kelly, payoff asymmetry, efficiency, liquidity, breakout, pullback and spike diagnostics. The central empirical question is whether these signals identify different danger states and whether their combination improves downside outcomes at matched average exposure.

The uploaded performance table shows that the previous “Advanced smart Bayesian” implementation was not merely conservative; it was malfunctioning or severely miscalibrated. It produced 0.71% annualized return, a -0.08 Sharpe ratio and a -27.62% maximum drawdown, while the simpler combined managed overlay achieved a 0.87 Sharpe and -7.31% maximum drawdown. The audit identified several mechanisms capable of producing exactly that failure: pure CVaR minimization rather than return maximization under a risk budget, one-day scenarios used for a monthly decision, smart cash exposure renormalized back to 100%, overly harsh multiplication of several timers, and next-month turnover computed from target rather than executed weights.

Deliverable	Status
Updated code package	Console-first pipeline with corrected scenario units, CVaR budget, stable walk-forward selection, smart timers, train/test metrics, risk-matrix export and Plotly dashboard.
Journal blueprint	Full section-by-section OPRE article plan, mathematical model, propositions, hypotheses, benchmarks, ablations and robustness checks.
Code validation	Repository compiles; four core tests pass; synthetic reporting smoke test generates all expected artifacts.
Historical results	Not regenerated because the raw point-in-time ETF dataset is absent. Any numerical claim must be rerun locally or on the research server.
“Advanced smart Bayesian must be best”	Rejected as a coding objective. The correct objective is to be best or near-best under frozen out-of-sample promotion criteria without data leakage.

Contents and Recommended Manuscript Flow

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1. Decision Problem and Operations Research Positioning

1.1 State, action, uncertainty and objective

At monthly rebalance date t , the decision maker observes an information set F_t containing only prices, volumes, realized returns, prior model states, the previous executed portfolio, and the point-in-time ETF master available by t . The action has two parts: a long-only composition vector w_t and a total risky-exposure fraction k_t . The remaining fraction $1-k_t$ is held in cash or a short-duration risk-free proxy. The next holding-period return is unknown and represented by a scenario matrix generated from F_t .

$$S_t = (F_t, A_t, w_{t-1}^{\text{exec}}, B_t); \quad a_t = (w_t^{\text{target}}, \alpha_t, k_t)$$

$$w_t^{\text{exec}} = w_{t-1}^{\text{exec}} + \alpha_t (k_t w_t^{\text{target}} - w_{t-1}^{\text{exec}}), \quad 0 \leq \alpha_t, k_t \leq 1$$

The paper should define a policy π that maps the state into a feasible action, rather than describe disconnected scripts. The policy must be causal, reproducible and auditable. This framing makes the contribution suitable for Operations Research: it is a sequential decision system under estimation error, changing feasibility, coherent tail-risk constraints and transaction costs.

1.2 Why Operations Research is a plausible target

The official Operations Research editorial statement emphasizes analytical methods that improve decision making and expects methodological/mathematical and empirical rigor. The journal also recognizes contributions demonstrated through mathematics, data, software or computation. The current editorial structure explicitly includes Financial Engineering, Optimization and Data/Software/Computation areas. The official submission guidance states that a regular manuscript is normally limited to 30 pages excluding references, with an electronic companion that should not exceed the manuscript length. These constraints imply that the final paper must be compact: the core model, main propositions and decisive experiments belong in the manuscript; extended proofs, code maps, additional robustness tables and dashboards belong in the electronic companion.

OPRE expectation	How this project should satisfy it
Clear decision problem	Define F_t , eligible universe A_t , scenario fan R_t , prior executed weights, target/executed weights, cash exposure and cost.
Analytical contribution	Prove causality, scale invariance of ranks, convexity/LP representation of CVaR, feasibility conditions, turnover bounds, overlay boundedness and timer complementarity.
Algorithmic clarity	Provide one monthly walk-forward algorithm and exact pseudocode for universe, scenarios, optimization, execution, signal timers and selection.
Empirical rigor	Use frozen walk-forward tests, matched-exposure comparisons, strong benchmarks, cost sensitivity, bootstrap, PSR/DSR and failure-case analysis.
Computational reproducibility	Release configs, seeds, environment, trial ledger, output schema, dashboard and one-command reproduction script.

1.3 Recommended title and core claim

Recommended title: “Risk-Signal Diversification in Dynamic ETF Allocation: A Model-Consistent CVaR Timing Policy with Transaction Costs.”

Recommended claim: We develop a monthly long-only ETF decision policy with a causal changing universe, horizon-consistent empirical-Bayes scenarios, a return-maximizing allocation subject to a posterior CVaR budget, partial execution and a bounded exposure controller. The controller combines a backward realized-volatility signal with a forward model-CVaR signal derived from the same scenario fan used by the optimizer. The contribution is not that CVaR, Kelly, GARCH or momentum are individually new. The contribution is the model-consistent integration, the allocation-versus-timing decomposition, and evidence that signals with different information provenance warn in different states and improve downside outcomes out of sample.

2. Uploaded Materials and Code-State Comparison

The uploaded package already contained a substantial implementation layer described in the earlier 13-page article blueprint and three-page implementation report: smart signals, train/test reporting, a Plotly dashboard and notebook-to-console migration. The current audit confirmed that these ideas were present, but also found that several central claims in the earlier

report were not fully true in the executable path. Most importantly, the nested smart cash overlay was renormalized away, actual model matrices were not exported, strategy_run did not use the nested engine, and the scenario horizon was inconsistent with monthly execution.

Area	Earlier package / report	Audited v2 state
ETF universe	Rich metric set with EMA or rolling score smoothing.	Observation leakage fixed; pairwise beta fixed; Sharpe-like units fixed; top-N selection flag clarified; universe turnover/Jaccard diagnostics added.
Bayesian model	Scenario fan generated from posterior mean and covariance.	Scenario fan is now explicitly H-day; reproducible RNG; empirical-Bayes mean shrinkage; t shocks; actual daily/horizon covariance and correlation snapshots.
CVaR optimizer	CVaR tasks including pure CVaR and mixed objectives.	Primary policy is max expected return subject to an exact Rockafellar–Uryasev CVaR LP budget; max-weight feasibility and post-LP distortion fixed.
Smart signals	Kelly, risk-reward, GARCH proxy, ER, ATR regime, liquidity, breakout, spike, pullback, net move.	Signals separated into asset tilt, forward conviction and portfolio timers; explicit backward-vol and forward-CVaR timers; default geometric combination prevents exposure collapse.
Train/test	Nested engine existed in main optimizer runner.	Stable validation/internal-test score with degradation penalty; optimizer-only and strategy-only experiments separated; strategy runner integrated.
Execution state	Target weights saved and carried forward.	Actual executed weights are now carried forward when $\alpha < 1$ and stored alongside target weights.
Dashboard	Interactive Plotly report with ex-post risk matrix.	Actual model covariance/correlation JSON, risky exposure, executed turnover, min/max weight filters, KPI cards and correct log-return benchmark conversion.
Metrics	Extended metrics appended to legacy stats.	Stats rewritten with guarded numerics, true benchmark beta/Treynor, PSR/DSR, rolling 6/12/24-month ratios and train/test export.

3. Diagnosis of the Reported Strategy Table

The table below reproduces the key values supplied with FINAL_results.ipynb. These values are treated as diagnostic evidence from the old implementation, not as validated results from the corrected code.

Variant	Ann. %	Vol. %	Sharpe	Sortino	MaxDD %	Calmar	CVaR95 %	PSR %	DSR %
Static (no overlay)	9.74	13.43	0.61	0.99	-19.34	0.50	-6.58	94.21	34.26
Dynamic base	11.27	13.70	0.70	1.13	-17.64	0.64	-6.93	96.26	43.41
+ Realized-vol overlay	10.48	10.35	0.82	1.61	-11.61	0.90	-4.38	98.45	55.67
+ Model-CVaR overlay	9.76	10.76	0.74	1.16	-11.97	0.82	-5.75	96.78	46.62
+ Combined (managed)	9.26	8.33	0.87	1.74	-7.31	1.27	-3.55	98.91	60.21
+ Advanced smart	0.71	9.73	-0.08	-0.14	-27.62	0.03	-7.16	41.21	1.16

Variant	Ann. %	Vol. %	Sharpe	Sortino	MaxDD %	Calmar	CVaR95 %	PSR %	DSR %
Bayesian									
Benchmark 1/N	10.41	14.44	0.62	0.81	-21.36	0.49	-8.71	93.71	35.79
Markowitz max-Sharpe	3.40	5.57	0.27	0.26	-15.45	0.22	-3.85	75.03	10.56
Markowitz min-variance	1.99	1.68	-0.01	-0.00	-5.18	0.38	-1.30	49.18	2.08

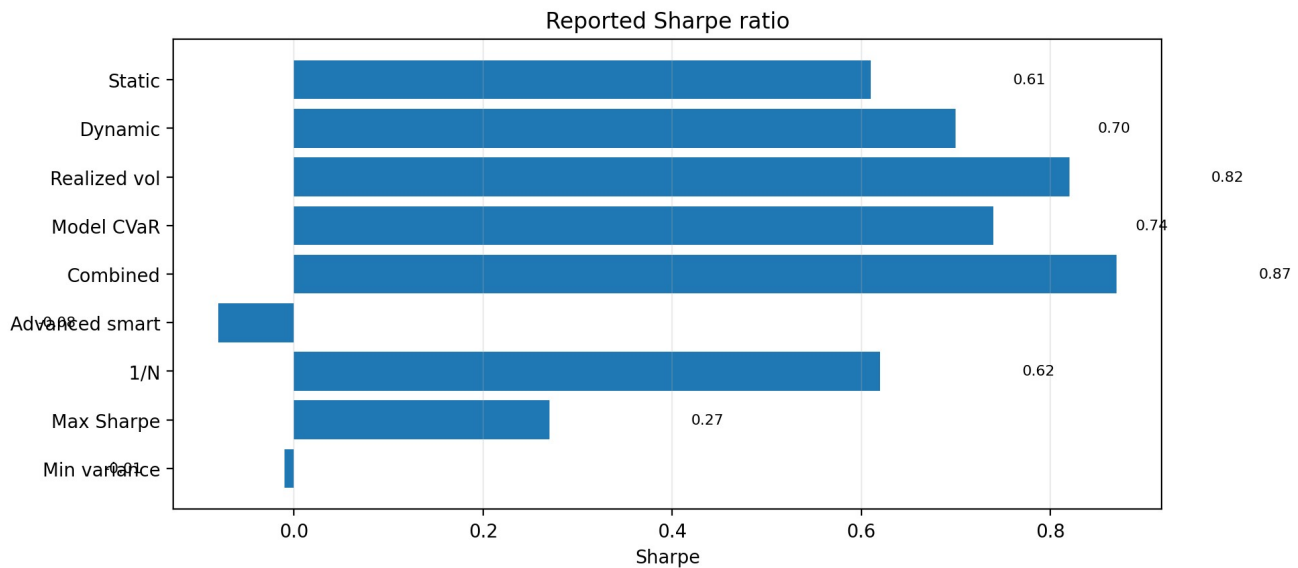


Figure 1. Reported Sharpe ratios from the old notebook output. The Advanced smart variant is the only clearly negative strategy.

The combined managed overlay is the strongest old result on downside-adjusted measures: Sharpe 0.87, Sortino 1.74, maximum drawdown -7.31%, Calmar 1.27 and CVaR -3.55%. Its annualized return is lower than the dynamic base, which is exactly the trade-off expected from holding cash in risky periods. The Advanced smart variant, however, loses both return and protection. A -27.62% maximum drawdown with only 9.73% volatility indicates that the strategy did not merely lower exposure; it allocated or timed risk badly.

3.1 Most likely failure mechanisms

- Pure CVaR objective: the advanced YAML selected `task_type=cvar_returns`. Minimizing tail loss alone can concentrate in assets with low forecast return and does not solve the intended return-risk decision problem.
- Horizon mismatch: the old scenario engine generated one-step outcomes while the optimization, execution and forward-risk interpretation were monthly. Means, volatilities and CVaR budgets were therefore not commensurate.
- Cash-overlay normalization bug: smart weights were multiplied by an exposure fraction and then normalized back to one in `build_full_weight_series`. The intended cash allocation disappeared.
- Overmultiplication: Kelly, risk-reward, volatility and quality scales were multiplied. Four moderately cautious values such as 0.7 produce 0.24, causing unstable exposure cliffs and then a hard floor.
- Incorrect portfolio state: `w_current` was set equal to the target even when partial execution used $\alpha < 1$. The next rebalance calculated turnover from a portfolio that had never been held.
- Unstable posterior mean: the old configuration used a strong momentum tilt of 0.30 and random horseshoe behavior, which can make the expected-return vector dominate a constrained optimizer.
- Selection leakage/instability risk: legacy alpha grid-search functions choose the best realized in-sample period and some legacy runners compare many methods without a complete trial ledger.
- Risk audit mismatch: the dashboard displayed an ex-post covariance from realized outputs, not the matrix used by the model at the decision date.

3.2 What “make Advanced smart Bayesian best” should mean

It is scientifically invalid to edit the code until a named method wins the same full-sample table. The correct requirement is a promotion gate. The advanced policy becomes the champion only when it beats predeclared baselines on a primary out-of-

sample score, does not violate drawdown/CVaR constraints, remains stable across windows and costs, and retains acceptable average risky exposure. If it fails, the fallback policy remains the champion. This turns a desired outcome into a falsifiable model-development protocol.

4. Corrected Research Architecture

Monthly causal decision policy and audit trail

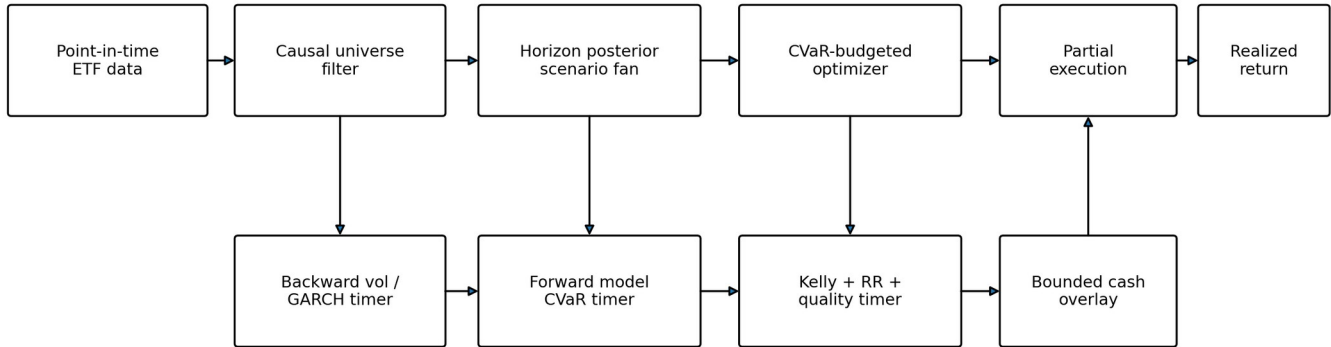


Figure 2. Corrected monthly decision architecture. Every arrow is causal and every major object is exported for audit.

27. At rebalance t , construct the eligible set A_t from point-in-time prices, volumes and prior smoothed scores.
28. Estimate a shrunk daily mean vector and covariance matrix from data dated no later than t .
29. Generate S holding-period scenarios by simulating H daily shocks and summing log returns or compounding simple returns.
30. Generate a compact candidate set of CVaR-budgeted or mixed-objective portfolios and partial-execution fractions.
31. Rank candidate recipes on a validation window, retest only the top recipes on a later internal-test window, and penalize validation-to-test degradation.
32. Refit the winning recipe on all history known at t ; do not use the next month.
33. Apply bounded asset-level tilts from signal quality, Kelly and payoff asymmetry.
34. Compute backward volatility and forward posterior-CVaR timers, combine them with the conviction timer, and preserve the resulting cash allocation.
35. Execute only α_t of the change, charge turnover cost, record actual executed weights, and evaluate the untouched next month.
36. Export PnL, target/executed weights, scenario risk, matrices, selection tables, smart-signal diagnostics and an interactive dashboard.

5. Point-in-Time ETF Universe Construction

5.1 Why the universe is part of the optimization model

The eligible universe is not a preprocessing detail. It defines the feasible action set and can materially change portfolio results. Optimizers amplify estimation error: a noisy expected return or covariance entry can receive a large weight when the eligible set contains illiquid, newly listed or pathologically unstable ETFs. The filter should therefore be presented as an operational feasibility and estimation-risk control, not as a proprietary alpha signal.

$$A_t = \{ i : H_{i,t} \geq H_{min}, P_{i,t} \geq P_{min}, V_{i,t} \geq V_{min}, Q_{i,t} = 1 \}$$

Here H is the number of actual observations, P is price, V is liquidity and Q aggregates optional benchmark-relative safeguards. Every term is F_t -measurable. The v2 code counts observations before filling and never backward-fills pre-inception prices.

5.2 Monthly metric definitions

Metric	Formula	Economic role
Price change	$P_{i,end} / P_{i,start} - 1$	Short-horizon momentum and demand.

Metric	Formula	Economic role
Benchmark excess	$r_i - r_{SPY}$	Separates ETF strength from broad equity beta.
Annualized risk-adjusted return	$\text{mean}(r_{i,d}) / \text{sd}(r_{i,d}) \times \sqrt{252}$	Rewards cleaner paths rather than raw return only.
Risk-adjusted excess	$(r_i - r_{SPY}) / \sigma_i$	Relative strength normalized by own risk.
Positive-day ratio	$\#\{r_{i,d} > 0\} / \#\{\text{valid days}\}$	Persistence rather than one large jump.
Total/dollar volume	$\Sigma \text{ volume}; \Sigma \text{ price} \times \text{volume}$	Tradability and rough capacity.
Volume change	$(V_t - V_{t-1}) / V_{t-1}$	Liquidity improvement/deterioration.
Volatility/downside volatility	$\text{sd}(r) \times \sqrt{252}; \text{sd}(\min(r, 0)) \times \sqrt{252}$	Symmetric and downside risk.
Maximum drawdown	$\min_s \text{Wealth}_s / \max_{\{u \leq s\}} \text{Wealth}_u - 1$	Path-dependent loss.
Amihud illiquidity	$\text{mean}(r_d / \text{dollar volume}_d)$	Daily price-impact proxy.
Beta/correlation to SPY	$\text{pairwise cov}(r_{i,m}) / \text{var}(r_m); \text{corr}$	Market sensitivity diagnostics.
Momentum ratio	$(1 + r_i) / (1 + r_{SPY}) - 1$	Scale-free relative strength.

5.3 Cross-sectional rank aggregation

$$\text{Score}_{\{i,t\}} = [\Sigma_j a_j \text{RankHigh}(x_{\{i,j,t\}}) + \Sigma_k b_k \text{RankLow}(z_{\{i,k,t\}})] / [\Sigma_j a_j + \Sigma_k b_k]$$

Percentile ranks remove units and reduce the influence of extreme scales. They do not eliminate model risk: metric weights, lookback windows and the hard benchmark filter remain design choices. The paper should therefore compare the full score with pure liquidity, fixed-universe, no-benchmark-filter and randomized-liquid-universe baselines.

5.4 EMA and rolling-score variants

$$S^{\{EMA\}}_{\{i,t\}} = \lambda S^{\{raw\}}_{\{i,t\}} + (1-\lambda) S^{\{EMA\}}_{\{i,t-1\}}, \quad 0 < \lambda \leq 1$$

$$S^{\{roll\}}_{\{i,t\}} = (1/L) \Sigma_{\{\ell=0\}}^{L-1} S^{\{raw\}}_{\{i,t-\ell\}}$$

monthly_etf_filtration_2.py is the primary EMA policy. monthly_etf_filtration.py is now a clean robustness variant that imports the same causal metric engine and averages scores across prior rebalance months. The output also reports half-L1 universe turnover and Jaccard overlap. This makes universe stability an observable outcome rather than an undocumented side effect.

5.5 Recommended universe robustness table

Variant	Purpose
Fixed high-liquidity ETF list	Separates dynamic-universe value from allocation/timing value.
Top dollar volume only	Tests whether the composite screen beats a transparent implementability screen.
No SPY filter	Measures the contribution and possible bull-market bias of relative-strength gating.
Top 50 / 100 / 150 / 220	Tests dimension and capacity sensitivity.
EMA $\lambda = 0.25 / 0.50 / 0.75 / 1.00$	Tests universe inertia and turnover.
Rolling score $L = 1 / 3 / 6$ months	Alternative smoothing policy.
Random top-N from liquid set	Placebo for selection alpha.
Point-in-time master with delisted/merged ETFs	Required survivorship-bias control.

6. Posterior-Predictive Scenario Engine and Covariance Estimation

6.1 Unit-consistent holding-period scenarios

The corrected model treats historical inputs as daily returns and outputs H-day scenarios. For log returns, daily simulated outcomes are summed; for simple returns, they are compounded. This is essential because the CVaR budget, forward risk timer and realized monthly PnL must share the same unit.

$$r_{[s,h]} = \mu_t + L_t z_{[s,h]}, \quad z_{[s,h]} \sim \mathcal{N}(0, I) \text{ or } \mathcal{N}(0, I), \quad L_t L_t^T = \Sigma_t$$

$$R^{\wedge}\{(H)\}_{[s,t]} = \Sigma_{[h=1] \wedge \{H\}} r_{[s,h]} \quad (\text{log returns})$$

$$R^{\wedge}\{(H)\}_{[s,t]} = \Pi_{[h=1] \wedge \{H\}}(1 + r_{[s,h]}) - 1 \quad (\text{simple returns})$$

Under independent daily log shocks, $\text{Cov}(R^{\wedge}(H)) = H\Sigma$. With serial dependence, this equality is only approximate because autocovariances contribute. A stronger future version can simulate VAR/DCC-GARCH dynamics or block-bootstrap daily vectors.

6.2 Empirical-Bayes mean shrinkage

$$\hat{\mu}_{[i,t]} = \kappa_{[i,t]} \bar{\mu}_{[i,t]} + (1 - \kappa_{[i,t]}) \mu^{\wedge}\{\text{prior}\}_{[i,t]} + \delta \cdot \text{Momentum}^{\wedge}\{12-1\}_{[i,t]}$$

The corrected default is deterministic empirical-Bayes shrinkage for reproducibility. A moderate 12–1 momentum tilt is optional and clipped relative to daily volatility. Random horseshoe draws remain available only as an explicit experiment. The paper must not call this a fully Bayesian posterior unless pointwise likelihood, priors and posterior sampling are implemented and diagnosed. “Empirical-Bayes shrinkage scenario engine” is accurate wording for the current default.

6.3 Covariance alternatives

Estimator	Use in paper	Failure mode / test
Sample covariance	Transparent baseline.	Unstable when N/T is large.
Ledoit–Wolf shrinkage	Primary stable estimator.	May miss fast regime changes.
EW covariance	Recency-sensitive robustness.	Noisy during isolated shocks.
Marchenko–Pastur cleaning	Eigenvalue-noise control.	Threshold assumptions may be too stylized; use as a robustness check with $q=N/T$ and report sensitivity.
Forecast-vol diagonal rescaling	Inject GARCH/EW volatility into dependence structure.	Correlation assumed more stable than volatility.
DCC-GARCH or realized covariance	Future high-complexity extension.	Estimation and computational burden.
Distributionally robust covariance set	Strong OPRE extension.	Requires ambiguity-radius calibration.

6.4 Scenario diagnostics

The old code evaluated scenario rows as if they were a time-ordered forecast path. The v2 metrics instead compare the posterior distribution for each ETF with the realized H-day vector. Diagnostics include predictive log score, a clearly labeled predictive-deviance proxy, rank information coefficient, MAE, RMSE, interval coverage and sign precision. Full WAIC is not reported because the framework does not retain pointwise posterior log-likelihood draws.

7. CVaR-Budgeted Allocation, Turnover and Execution

7.1 Empirical CVaR

$$CVaR_{\beta}(w) = \min_{\eta} \{ \eta + [1/((1-\beta)S)] \Sigma_{[s=1] \wedge \{S\}} \max(-R_s^T w - \eta, 0) \}$$

CVaR is used as a coherent tail-risk budget, not as an alpha source. The previous advanced configuration minimized CVaR directly. The corrected primary policy maximizes posterior expected holding-period return subject to a CVaR budget tied to a transparent reference portfolio.

7.2 Exact return-CVaR linear program

$$\text{maximize}_w \quad \mu_t^T w - \lambda_t \|w - w_{[t-1]}\|_1$$

$$\text{subject to } CVaR_{\beta,t}(w) \leq c_t; \quad 1^T w = 1; \quad 0 \leq w_i \leq u_i \\ c_t = \kappa \cdot CVaR_{\beta,t}(w^{1/N})$$

For a fixed scenario matrix, auxiliary variables η and u_s linearize CVaR and positive/negative turnover variables linearize the L1 term. The solver uses SciPy/HiGHS. If $u_i < 1/N$, the full-investment constraints are infeasible; v2 enforces a feasible effective cap. The LP solution is no longer thresholded afterward, because hard minimum-weight cleanup can violate the tail-risk budget that the LP just enforced.

7.3 Partial execution and transaction costs

$$w_t^{exec} = w_{t-1}^{exec} + \alpha_t (w_t^{overlay} - w_{t-1}^{exec}) \\ Turnover_t = \|w_t^{exec} - w_{t-1}^{exec}\|_1; \quad Cost_t = c \cdot Turnover_t$$

The execution state must be w_{exec} , not w_{target} . When $\alpha < 1$, carrying the target forward understates or misstates next-month turnover. The corrected runners store `target_weights` and `executed_weights` separately and use the executed vector as the next state.

7.4 Alternative objectives as ablations

Objective	Role
Return subject to CVaR budget	Primary interpretable policy.
Mean–CVaR–Sharpe score	Nonlinear alternative; requires careful scaling of terms.
Pure minimum CVaR	Risk-only benchmark, not expected to maximize growth.
Minimum variance	Low-risk benchmark.
Maximum Sharpe	Estimation-error-sensitive benchmark.
1/N	Strong naive benchmark.
Hold current	Turnover-free candidate and stability fallback.

8. Risk-Signal Diversification and Smart Position Sizing

8.1 Three distinct layers

Layer	Signals	Action
Asset quality tilt	Efficiency ratio, breakout, liquidity, pullback, net move, ATR regime, normalized spike.	Gently tilt optimizer weights; do not replace the optimizer.
Forward conviction	Scenario Kelly and risk-reward ratio.	Bounded multiplier on selected risky-sleeve composition.
Portfolio risk timing	Backward realized/GARCH volatility and forward model-CVaR.	Scale total risky exposure; residual goes to cash/T-bills.

8.2 Asset-level formulas

$$ER_{i,t} = (P_{i,t} - P_{i,t-L}) / \sum_{j=1}^L (P_{i,t-j+1} - P_{i,t-j})$$

$$ATR_{regime}_{i,t} = ATR^{short}_{i,t} / ATR^{long}_{i,t}$$

$$Breakout_{i,t} = (P_{i,t} - Low_{i,t,L}) / (High_{i,t,L} - Low_{i,t,L})$$

$$Spike_{i,t} = r_{i,t} / \sigma_{i,t,L}; \quad Pullback_{i,t} = (High_{i,t,L} - P_{i,t}) / (High - Low)$$

When true OHLC is unavailable, the current code uses a price proxy reconstructed from returns and an absolute price-change proxy for ATR. The manuscript must call it an ATR proxy. With true OHLCV, replace it with standard true range and quoted-spread liquidity.

8.3 Kelly and payoff asymmetry

$$f^{Kelly}_{[i,t]} = clip(E_s[R_{[s,i]}] / Var_s[R_{[s,i]}], 0, f_{max})$$

$$RR_{[i,t]} = E_s[\max(R_{[s,i]}, 0)] / E_s[\max(-R_{[s,i]}, 0)]$$

Kelly is highly sensitive to mean-estimation error. It is therefore fractional, long-only, clipped and used only as a gentle multiplier. It must never be allowed to override diversification, caps or CVaR feasibility. A future robust Kelly extension can shrink the edge toward zero or optimize worst-case expected log growth over an ambiguity set.

8.4 Volatility clustering

$$h_{[t+1]} = \omega + a \varepsilon_t^2 + b h_t, \quad a+b < 1; \quad \hat{\sigma}^{(H)}_{[t+1]} = \sqrt{H h_{[t+1]}}$$

The code implements a fast fixed-parameter GARCH(1,1)-style recursion. It captures persistence without adding a heavy dependency, but it is not maximum-likelihood GARCH estimation. The empirical section should compare this proxy with realized, EW and optionally fitted GARCH forecasts.

8.5 Backward and forward exposure timers

$$k^{vol}_t = clip(\sigma^* / \max(\hat{\sigma}^{realized}_t, \hat{\sigma}^{GARCH}_t), k_{min}, 1)$$

$$k^{cvar}_t = clip(c^*_t / CVaR^{model}_{[\beta,t]}, k_{min}, 1)$$

$$k^{conv}_t = [k^{Kelly}_t k^{RR}_t k^{quality}_t]^{1/3}$$

$$k_t^{geom} = [k^{vol}_t k^{cvar}_t k^{conv}_t]^{1/3}$$

The original product of four controls was too aggressive. The corrected default is a geometric mean, which remains multiplicative in spirit but does not collapse exposure as quickly. Product, minimum and convex blend are explicit ablations. For the core paper contribution, report both the original two-signal product and a matched-exposure geometric/blended policy; this shows whether gains come from complementarity rather than simply holding more cash.

8.6 Signal-quality composite

$$Q_{[i,t]} = sigmoid(a_{ER} z(ER) + a_B z(Breakout) + a_L z(\log Liquidity) + a_R z(\log(1+RR)) + a_M z(NetMove) + a_P z(PullbackGood) - a_V z(ATRRegime) - a_S z(|Spike|))$$

Every coefficient is a hyperparameter and must be included in the trial ledger if tuned. A journal-quality design should either freeze economically motivated coefficients before the final test or estimate them only in the training layer with a low-dimensional regularized model.

9. Train/Validation/Internal-Test and Next-Month Evaluation

A single full-sample backtest cannot distinguish a useful policy from a selected lucky parameter set. The corrected engine partitions only data known at t into train, validation and internal test. The next month remains untouched. Candidate recipes are generated from train, ranked on validation, retested on internal test, then the selected recipe is refit on all history known at t and executed next month.

$$RobustScore_j = (1-\rho) V_j + \rho T_j - \gamma / |T_j - V_j|$$

V_j and T_j are validation and internal-test scores. The degradation penalty discourages candidates that look excellent only on validation. This is still model selection, so all candidates and all alpha values count toward multiple-testing correction.

Stage	Data use	Forbidden use
Train	Fit means, covariance, signals and candidate weights.	No validation, internal-test or next-month returns.
Validation	Rank the full candidate set.	No refit to validation outcome except recipe selection.
Internal test	Rerank only top validation candidates and assess degradation.	No use of the true next month.
Refit at t	Fit winning recipe on all data through t .	No $t+1$ observations.
External next month	Measure realized net return and update controller state.	Never used to revise the action already taken.

9.1 Recommended rolling protocol

Use at least two complementary protocols: (i) the monthly nested policy above, which mirrors live operation; and (ii) coarse chronological blocks such as train 2010–2017, validation 2018–2019, test 2020–2022, followed by a final untouched 2023–2025 holdout if the data allow. Exact dates depend on data availability. The final holdout must be opened only after the search grid, hypotheses and primary metric are frozen.

9.2 Champion–challenger promotion gate

37. Primary score: test-period Sharpe or utility score with a predeclared turnover penalty.
38. Risk gates: maximum drawdown and CVaR may not be worse than the current champion by more than specified tolerances.
39. Exposure gate: compare at matched average risky exposure and report cash fraction.
40. Stability gate: positive or acceptable results across most rolling folds, not only aggregate terminal wealth.
41. Cost gate: superiority survives at least 10 and 25 bps per unit turnover, with 50 bps stress.
42. Inference gate: paired bootstrap confidence interval and PSR/DSR are reported; DSR uses the full trial count.
43. Fallback: if the advanced policy fails, retain Combined Managed or another predeclared champion. Do not alter the final holdout.

10. Metrics, Inference and Multiple-Testing Control

10.1 Performance and downside metrics

$$Expectancy = p_{win} \cdot AvgWin + (1 - p_{win}) \cdot AvgLoss$$

$$ProfitFactor = \sum_{\{r_t > 0\}} r_t / |\sum_{\{r_t < 0\}} r_t|$$

$$AverageDrawdown = mean\{DD_t : DD_t < 0\}$$

$$RollingSharpe_{[t,L]} = \sqrt{12} \cdot mean(r - r_f/12) / sd(r - r_f/12)$$

$$RollingSortino_{[t,L]} = [12 \cdot mean(r) - r_f] / [\sqrt{12} \cdot \sqrt{mean(min(r, 0)^2)}]$$

The v2 stats module reports cumulative and annualized return, volatility, Sharpe, Sortino, maximum and average drawdown, Calmar, VaR, CVaR, downside deviation, win rate, average win/loss, expectancy, profit factor, Omega, pain ratio, MAD, skewness, excess kurtosis, PSR and DSR. Beta and Treynor are calculated only when a benchmark series exists; the old beta=1 shortcut was removed.

10.2 Probabilistic and Deflated Sharpe ratios

$$PSR(SR^*) = \Phi((SR - SR^*)\sqrt{(n-1)} / \sqrt{[1 - \gamma_3 SR^* + ((\gamma_4 - 1)/4) SR^{*2}]})$$

PSR estimates the probability that the true Sharpe exceeds a benchmark after accounting for sample length, skewness and kurtosis. DSR replaces the zero benchmark with the expected maximum Sharpe generated by the number and dispersion of tried strategies. The critical implementation requirement is a complete trial ledger. Counting only final winners materially overstates DSR.

10.3 Statistical inference

- Paired moving-block bootstrap of monthly return differences to preserve serial dependence.
- Confidence intervals for annualized return, Sharpe difference, maximum drawdown difference and expected shortfall difference.
- Forecast-validity regressions: next-month realized volatility/tail loss on model-CVaR and realized-vol signals.
- Warning-event contingency tables: months flagged by forward only, backward only, both and neither.
- Stress-event case studies with decision-date information and no ex-post reinterpretation.
- Superior Predictive Ability or Model Confidence Set tests as optional extensions when comparing many policies.

11. Mathematical Propositions and Proof Sketches

Proposition 1 — Causality of the universe. If every metric used at t is a deterministic function of observations dated no later than t and smoothing uses only scores through $t-1$, then A_t is F_t -measurable.

Proof sketch. Prices, volumes, observation masks and prior scores are F_t -measurable. Compositions of measurable deterministic functions and rank operators remain measurable; therefore the selected top-N set is known at t .

Proposition 2 — Rank-score invariance. A percentile rank is invariant under any strictly increasing transformation of a metric.

Proof sketch. Strictly increasing maps preserve all pairwise orderings. Therefore the empirical rank of each observation is unchanged, apart from tie conventions. This explains why rank aggregation reduces unit sensitivity, but not weight or lookback sensitivity.

Proposition 3 — EMA contraction. For $S_t = \lambda X_t + (1-\lambda)S_{t-1}$, $|S_t - S_{t-1}| = \lambda |X_t - S_{t-1}|$.

Proof sketch. Subtract S_{t-1} and factor λ . Thus λ directly bounds score movement relative to replacement by the raw score. Top-N membership turnover can still be discontinuous near rank cutoffs, so turnover must be measured empirically.

Proposition 4 — CVaR convexity and LP exactness. For fixed scenarios, empirical CVaR is convex in w and the return-maximization problem with an L1 turnover term and linear bounds has an exact linear-program representation.

Proof sketch. The pointwise maximum of affine losses is convex. Introducing η and nonnegative u_s with $u_s \geq -R_s^T w - \eta$ yields the Rockafellar–Uryasev representation. Positive/negative turnover variables linearize $|w - w_{\text{prev}}|$.

Proposition 5 — Feasibility of the reference CVaR budget. If equal weights satisfy the box constraints and $c_t \geq \text{CVaR}(w^{\{1/N\}})$, the feasible set is nonempty.

Proof sketch. The equal-weight portfolio is a witness. The max-weight requirement is $u \geq 1/N$. This is why the implementation raises an infeasible cap to at least $1/N$.

Proposition 6 — Partial-rebalance turnover bound. For $w_{\text{exec}} = w_{\text{prev}} + \alpha(w_{\text{target}} - w_{\text{prev}})$, $0 \leq \alpha \leq 1$, $\|w_{\text{exec}} - w_{\text{prev}}\|_1 = \alpha \|w_{\text{target}} - w_{\text{prev}}\|_1$.

Proof sketch. Direct substitution and positive homogeneity of the L1 norm. Partial execution cannot create more turnover than full execution.

Proposition 7 — Positive homogeneity of risky-sleeve CVaR. For $k \geq 0$ and a zero-return cash asset, $\text{CVaR}(k w) = k \text{CVaR}(w)$.

Proof sketch. CVaR is positively homogeneous. Therefore a correctly preserved cash overlay scales the risky sleeve's scenario loss linearly, ignoring deterministic cash translation. Renormalizing $k w$ back to one destroys this guarantee.

Proposition 8 — Bounded controller. If every timer lies in $[k_{\text{min}}, 1]$ and the combination is product, minimum, convex blend or geometric mean, the final exposure lies in a known bounded interval within $[0, 1]$.

Proof sketch. Products and geometric means of numbers in $[0, 1]$ remain in $[0, 1]$; minima and convex combinations do also. The explicit final clip enforces the lower bound.

Proposition 9 — Product-timer warning coverage. For $k_{\text{vol}}, k_{\text{cvar}} \in [0, 1]$, $k_{\text{vol}} k_{\text{cvar}} \leq \min(k_{\text{vol}}, k_{\text{cvar}})$.

Proof sketch. Because multiplying either factor by a number no greater than one cannot increase it. The product de-risks whenever either timer de-risks, at the cost of lower average exposure.

Proposition 10 — External next-month validity. If candidate generation, validation, internal testing and refitting use only F_t , then the realized return over $(t, t+1]$ is an out-of-sample observation conditional on the selected policy.

Proof sketch. The selected action is F_t -measurable. The next-period outcome is not used in the selection map, so it remains outside the training sigma-algebra. Repeated model development after observing the whole history still creates research-level selection bias, which DSR and a final frozen holdout address.

12. Empirical Design, Benchmarks, Ablations and Robustness

12.1 Primary research questions

Research question	Test	Required evidence
RQ1: Does model-CVaR forecast risk?	Regress next-month realized volatility, drawdown and tail loss on the decision-date model-CVaR.	Positive and stable risk relation; return-direction prediction is not required.
RQ2: Are forward and backward timers complementary?	Exposure correlation, warning overlap and lead/lag event study.	Weak-to-moderate correlation and months flagged by only one timer.
RQ3: Does the combination improve decisions?	Compare forward-only, backward-only, product, geometric mean and convex blend at matched exposure.	Best or near-best downside-adjusted OOS performance with cost robustness.
RQ4: Where does value come from?	Decompose universe, allocation, execution and timing increments.	Risk timing explains improvement rather than an opaque optimizer.
RQ5: Is the advanced layer stable?	Rolling folds, parameter-selection	No dependence on one window or one

Research question	Test	Required evidence
	frequency, degradation and failure cases.	signal coefficient.

12.2 Required benchmarks

- SPY buy-and-hold; 1/N on the same eligible universe; fixed-universe 1/N; minimum variance; maximum Sharpe; risk parity/inverse volatility; static Bayesian-CVaR allocation; dynamic base without overlay; backward-only timer; forward-only timer; combined timer; hold-current; cash-equivalent benchmark.

12.3 Mandatory ablation matrix

Block removed or changed	Question answered
No dynamic universe	Does the changing universe add value or create selection bias?
No expected-return tilt	Is performance driven by covariance/risk control?
Sample vs shrinkage vs cleaned covariance	How sensitive is allocation to matrix estimation?
No Kelly / no RR / no quality tilt	Which smart component helps or harms?
Product vs geometric mean vs convex blend	Is severe de-risking responsible for CAGR loss?
No partial execution / fixed α	Does alpha selection add value after costs?
No cash overlay	Does composition alone improve results?
Matched average exposure	Does the timer add state information beyond lower beta?
Alternative CVaR level/budget	Is tail-risk control robust to β and κ ?

12.4 Transaction-cost and capacity design

Run cost sensitivity at 0, 5, 10, 25 and 50 basis points per unit of one-way L1 turnover. The current `c_bps` variable is a decimal cost coefficient: 0.001 corresponds to 10 basis points on 100% turnover. Rename or document this unit in the final code. For capacity, report selected ETFs' rolling median dollar volume, maximum portfolio weight relative to liquidity, and an Amihud-based stress cost. Daily close/volume cannot support a claim of institutional implementation without stronger spread/impact data.

13. Defensible Novelty Versus Known Methods

Component	Prior status	Safe contribution wording
Markowitz / shrinkage / random-matrix cleaning	Known.	Stabilization tools within the decision engine; not inventions.
CVaR and Rockafellar–Uryasev LP	Known.	A coherent scenario tail-risk budget tied to a transparent reference book.
Volatility-managed exposure	Known.	Canonical backward comparator and one input to the combined policy.
GARCH volatility clustering	Known.	Fast risk-regime proxy; not a novel volatility model.
Kelly criterion and risk-reward	Known.	Bounded conviction multipliers subordinate to portfolio constraints.
Momentum, liquidity, drawdown, Amihud, breakout/ER	Known features.	Causal operational universe/quality layer.
Forward posterior-CVaR timer from the optimizer's own scenario fan	Defensible integration contribution.	A model-consistent risk timer using the same uncertainty representation as allocation.
Backward + forward warning-source diversification	Defensible if evidence supports.	Risk-signal diversification: signals with different provenance cover different danger states.
Allocation-versus-timing decomposition	Strong empirical contribution.	Shows whether the optimizer selects alpha or the policy transforms risk.
Stable nested recipe selection plus full audit artifacts	Computational contribution.	Reproducible decision policy that controls parameter instability and exposes each monthly

Component	Prior status	Safe contribution wording
		action.

13.1 Stronger future OPRE extensions

- Distributionally robust CVaR: optimize against a Wasserstein or moment-based ambiguity set calibrated from scenario forecast errors.
- Endogenous signal combination: choose timer weights by minimizing a proper loss for future tail-risk forecasts, with coefficients constrained and estimated only in training.
- Multi-period stochastic programming: include turnover and exposure decisions across several future rebalances rather than myopic one-month optimization.
- Regime-switching posterior: jointly model covariance, liquidity and tail dependence with a hidden Markov or Bayesian change-point state.
- Robust Kelly: maximize worst-case expected log growth under mean/covariance ambiguity, then embed the result as a bounded sleeve scaler.
- Capacity-aware optimization: incorporate spread/impact estimates and liquidity constraints directly rather than only screening before optimization.

14. OPRE Manuscript Section-by-Section Blueprint

Section	Content and target length
Abstract (≤ 200 words)	Decision problem; method; model-consistent forward timer; complementary backward timer; main OOS risk result; no equations or unsupported alpha claim.
1. Introduction (3–4 pages)	Changing ETF feasibility, estimation error, tail risk, transaction costs; why allocation and exposure timing must use a consistent uncertainty model; contributions and findings.
2. Literature (3 pages)	Portfolio estimation error/robust optimization; CVaR/expected shortfall; volatility management; ES forecasting; Bayesian shrinkage; backtest selection and multiple testing.
3. Decision model (5 pages)	Information set, universe, scenarios, CVaR LP, execution, cash exposure and timer definitions.
4. Policy and algorithm (3 pages)	Monthly pseudocode, state transitions, nested selection, computational complexity and implementation details.
5. Theory (3 pages)	Causality, rank invariance, LP exactness, feasibility, turnover, homogeneity, bounded timers and complementarity. Longer proofs in companion.
6. Data and universe (3 pages)	Sources, ETF master, survivorship, missing data, metric formulas, filters, turnover and descriptive statistics.
7. Experimental design (3 pages)	Benchmarks, costs, frozen grid, folds, primary metric, bootstrap, PSR/DSR and matched-exposure design.
8. Results (5 pages)	Forecast validity $\rightarrow$ timer complementarity $\rightarrow$ policy performance $\rightarrow$ decomposition $\rightarrow$ failure cases.
9. Robustness (2–3 pages)	Universes, covariance, costs, timer forms, CVaR levels, subperiods and stress periods.
10. Conclusion (1 page)	Decision-policy contribution and honest limitations.
Electronic companion	Full code map, configs, trial ledger, extra tables/plots, proofs, dashboard and reproduction instructions.

14.1 Suggested abstract draft

We study a monthly long-only ETF allocation problem with a changing investable universe, noisy return and covariance estimates, tail-risk constraints, transaction costs, and time-varying total exposure. At each rebalance, a causal liquidity- and history-aware screen defines the feasible ETF set. An empirical-Bayes scenario engine generates holding-period return scenarios, and the portfolio maximizes posterior expected return subject to a scenario CVaR budget and turnover cost. The same scenario fan produces a forward model-CVaR forecast for the selected book. We combine this forward signal with a

backward realized-volatility signal and a bounded conviction signal, creating a risk-signal-diversification policy. A nested walk-forward procedure selects portfolio recipes using validation and internal-test data while preserving the next month for external evaluation. The empirical analysis tests whether the signals forecast different risk states and whether their combination improves downside outcomes at matched average exposure. The framework emphasizes transparent risk transformation and reproducible decision making rather than an unconditional claim of security-selection alpha.

14.2 Suggested contribution paragraph

44. We formulate ETF allocation and exposure as one causal stochastic decision policy with a point-in-time changing action set.
45. We introduce a model-consistent forward tail-risk timer derived from the same posterior scenario fan that defines the allocation CVaR constraint.
46. We formalize risk-signal diversification by combining forward model-CVaR with backward realized-volatility information and test warning-source complementarity at matched exposure.
47. We separate allocation, execution and timing contributions through a complete ablation and champion–challenger design.
48. We provide an auditable implementation that exports actual decision-date risk matrices, target/executed weights, trial-selection tables and interactive monthly diagnostics.

15. Interactive Dashboard and Reproducibility Layer

The self-contained Plotly HTML report is an audit artifact, not merely a presentation. It reads the same files produced by the strategy. The v2 dashboard contains KPI cards; strategy and benchmark growth/drawdown; overall/train/test metrics; rolling Sharpe/Sortino; selectable return histograms; executed turnover and risky exposure; eligible and selected ETF heatmaps; actual model covariance/correlation matrices by date; forward risk timers; dynamic parameters; selection audits; and monthly tables with both minimum and maximum weight filters.

Artifact	Purpose
pnl.csv	Net balance, PnL, cost, return and alpha.
real.csv / preds.csv	Realized and posterior-mean ETF holding-period returns.
weights.xlsx	Eligible flags, target weights, executed weights and retained compatibility weight column.
risk_matrices.json	Actual model daily/horizon covariance, correlation and posterior means.
forecast_risk.csv	Model CVaR/volatility, timers, overlay and risky exposure.
selection_audit.csv	Selected recipe, parameters, validation/internal-test evidence and robust score.
smart_rebalance_audit.csv	Portfolio timers, Kelly, RR, quality and per-ticker signal JSON.
train_test_metrics.csv / rolling_metrics.csv	Stable comparison and rolling diagnostics.
run_metadata.json	Return units, horizon, model/optimizer settings and warnings.
interactive_report.html	Portable one-file audit dashboard.

The included sample dashboard is generated from deterministic synthetic data and is labelled as a smoke test. It proves that the reporting pipeline works; it is not evidence about the research strategy.

16. Code Map, Commands and Performance Engineering

File	Research role
scripts/dataloader/monthly_etf_filtration_2.py	Primary EMA-smoothed causal ETF universe.
scripts/dataloader/monthly_etf_filtration.py	Rolling-score universe robustness variant.
scripts/models/bayessian.py	H-day empirical-Bayes posterior scenario engine and matrix snapshots.
scripts/optimizers/base.py	CVaR LP, turnover variables, feasible bounds and postprocessing.
scripts/optimizers/bayessian.py	Bayesian/CVaR objectives and risk calculations.
scripts/backtesting/walk_forward_engine.py	Candidate generation, validation/internal-test ranking and refit.

File	Research role
scripts/backtesting/engine_integration.py	Shared bridge for optimizer and strategy runners plus smart layer.
scripts/risk_controls/smart_signals.py	Asset quality, Kelly/RR, GARCH proxy, backward/forward timers and cash overlay.
scripts/runs/run.py	Primary optimizer runner; actual state, matrices and audit outputs.
scripts/runs/strategy_run.py	Strategy-only champion/challenger runner using the same nested engine.
scripts/results/stats.py	Robust metrics, PSR/DSR, train/test and rolling exports.
scripts/results/interactive_report.py	Self-contained Plotly audit dashboard.
scripts/runs/synthetic_smoke.py	Deterministic reporting smoke test; no research claims.
tests/test_core_pipeline.py	Horizon, reproducibility, CVaR, cash-overlay and universe-inception tests.

16.1 Console commands

```
python -m pip install -r requirements.txt
python -m pytest -q

python -m scripts.dataloader.monthly_etf_filtration_2 \
  --prices datasets/csv/NewClosePrice.csv \
  --volumes datasets/csv/Volume.csv \
  --business-dates datasets/excel/business_dates.xlsx \
  --top-n 100 --benchmark SPY --spy-filter smart \
  --lookback-days 90 --score-ema-alpha 0.5

python -m scripts.runs.run -c advanced_smart_bayesian.yaml
python -m scripts.runs.strategy_run -c strategy_smart.yaml
python -m scripts.results.interactive_report \
  --result-folder results/advanced_smart_bayesian
python -m scripts.runs.synthetic_smoke
```

16.2 Speed improvements already implemented

- Vectorized H-day scenario generation instead of Python loops over scenarios.
- One scenario fan per rebalance is reused by candidate optimizers and forward signals.
- Exact HiGHS linear programming for the primary return-CVaR problem.
- Compact parameter grid and top-K validation filtering before internal test.
- Shared postprocessing instead of repeated notebook/report logic.
- Console entry points remove notebook-only state and make profiling/reproduction possible.

16.3 Further speed improvements

- Cache rolling return windows, means, covariance matrices and eligibility masks by rebalance date.
- Persist input matrices as Parquet rather than repeatedly reading wide CSV/Excel files.
- Parallelize independent candidate evaluation after the shared scenario fan is built.
- Use successive halving or Bayesian optimization inside training instead of a large Cartesian grid.
- Warm-start nonlinear objectives from equal weight, previous weight and the LP solution.
- Profile with py-spy/cProfile before adding JAX; remove unused JAX imports from legacy runners.
- Keep the legacy combined/new runners only for reproduction; route new research through the unified engine.

17. Remaining Limitations and Required Final Validation

49. Run the corrected code on the exact raw dataset and regenerate every table and figure. The current package cannot establish performance without those files.
50. Build a point-in-time ETF master with inception dates, delistings, closures, mergers and ticker changes. A current surviving list creates survivorship bias.

51. Freeze the model universe, parameter grid, random seeds, primary score, risk gates and final holdout before the last run.
52. Log every candidate and research variant for DSR. The number of selected winners is not the number of trials.
53. Calibrate transaction costs from quoted spreads or run the full stress grid. Daily volume and Amihud are only proxies.
54. Compare policies at matched average risky exposure and with a cash/T-bill return, not zero cash return alone.
55. Use proper forward expected-shortfall scoring or forecast-loss functions when estimating timer combinations.
56. Report failure months and turnover/exposure path, not only aggregate Sharpe.
57. Check optimizer feasibility, realized CVaR breaches, concentration, matrix condition number and scenario calibration each month.
58. Only promote Advanced Smart Bayesian after it satisfies the predeclared champion–challenger gate; otherwise report that the simpler combined policy remains superior.

17.1 Final result tables required

- Main OOS performance table with SPY, 1/N, Markowitz, static, dynamic and all overlay ablations.
- Matched-exposure performance table.
- Forward-risk forecast-validity table.
- Timer complementarity/overlap table.
- Allocation-versus-timing decomposition.
- Train/validation/internal-test/final-holdout stability table.
- Universe turnover and composition diagnostics.
- Cost, universe-size, covariance and CVaR-parameter robustness tables.
- Bootstrap, PSR, DSR and trial-ledger table.
- Failure-case and stress-period table.

18. Conclusion

The project can become a credible Operations Research submission, but only if the paper and code are disciplined around the same decision problem. The most promising contribution is not a collection of indicators. It is a causal monthly policy in which universe selection, scenario generation, coherent-risk allocation, partial execution and total exposure are auditable parts of one system. The forward tail-risk signal is especially attractive because it is derived from the optimizer's own uncertainty model; combining it with a backward realized-volatility signal creates a clear hypothesis about information-source diversification.

The old Advanced smart Bayesian result should be treated as a failed prototype, not hidden or forced to win. The code corrections address concrete reasons for its failure. The final paper becomes stronger if the corrected advanced policy genuinely wins out of sample, but it remains publishable in a different form if the study shows that a simpler combined overlay is more robust. A rigorous negative or decomposition result is more credible than an optimized full-sample victory.

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Appendix A. Audit Validation Summary

Check	Result
Repository compilation	Passed with python -m compileall -q scripts.
Core tests	4 passed: scenario horizon/reproducibility, CVaR LP feasibility, cash-overlay preservation, ETF inception observation handling.
Synthetic reporting smoke test	Passed; generated PnL, weights, matrices, train/test metrics, rolling metrics and self-contained HTML.
Real ETF historical run	Not executed: raw datasets are not present in the uploaded archive.
Numerical performance claim	None made for the corrected policy. The supplied old table is analyzed only as diagnostic evidence.

Appendix B. Minimal Monthly Algorithm

Input: point-in-time prices/volumes, prior executed weights, cash balance

For each rebalance date t :

1. Build A_t using data dated $\leq t$; save universe diagnostics.
2. Split known history into train / validation / internal test.
3. Fit the H-day posterior scenario engine on train.
4. Generate optimizer/strategy candidates and alpha candidates.
5. Rank all candidates on validation; retain top K.
6. Rerank top K on internal test with degradation penalty.
7. Refit winning recipe on all data through t .
8. Compute asset quality, fractional Kelly and risk-reward tilts.
9. Compute backward vol timer and forward model-CVaR timer.
10. Combine timers; preserve cash exposure $k_t < 1$ where indicated.
11. Execute α_t of the target change and charge turnover cost.
12. Save target/executed weights, matrices, scenarios, diagnostics and audits.
13. Observe next-month realized return; update state.

Output: causal OOS return series and complete decision audit.